A Learning-Controlled Elephant Random Walk:

Martingale Structure, Diffusive Limits and Reproducible Evidence

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Abstract

The elephant random walk is a non-Markovian stochastic process whose next increment is drawn from its complete history and is then repeated or reversed. This article introduces a learning-controlled elephant random walk (LC-ERW), in which the memory parameter is no longer fixed but is selected online by a predictable learning rule. The construction is intended as a mathematically tractable bridge between long-memory random walks and artificial-intelligence mechanisms for adaptive exploration. The main result is a martingale normalisation valid for arbitrary predictable memory policies. Under a stable learning condition, namely convergence of the learned memory coefficient to a diffusive limit smaller than one half, the LC-ERW satisfies a central limit theorem with the same variance form as an elephant random walk with the limiting memory coefficient. The proof is based on triangular martingale convergence and explicit control of the predictable quadratic variation. Reproducible R experiments are provided to check the variance prediction and to illustrate the transition from adaptive behaviour to the limiting Gaussian regime.

Keywords: elephant random walk; artificial intelligence; online learning; martingales; central limit theorem; non-Markovian random walk.

1 Introduction

Random walks with memory provide simple models in which local moves are affected by a long historical record. The elephant random walk (ERW), introduced by Schütz and Trimper (2004), is a canonical example: at each time the walker selects one of its past increments and either repeats or reverses it. This mechanism generates diffusive, critical and superdiffusive regimes, depending on the memory parameter. Subsequent mathematical work has developed central limit theorems, martingale methods, urn connections and variants with restricted memory, random step sizes and delays (Coletti et al., 2017; Bercu, 2018; Baur and Bertoin, 2016; Gut and Stadtmüller, 2021, 2023; Dedecker et al., 2023; Aguech, 2024b,a).

Artificial intelligence (AI), particularly online learning and reinforcement learning, is also concerned with sequential decisions under uncertainty. In many algorithms, exploration is controlled by a parameter that is updated from the past. Yet the usual random-walk component of such methods is often Markovian or externally scheduled. This paper studies a complementary question: what happens when the memory parameter of an ERW is itself produced by a learning rule?

The contribution is a learning-controlled elephant random walk (LC-ERW). The walker has complete elephant memory, but the probability of repeating a recalled increment is chosen from the past by an arbitrary predictable learning algorithm. This includes deterministic schedules, online estimators, stochastic approximation recursions and bounded policy maps. The model is deliberately minimal: it is rich enough to represent AI-style adaptation, but constrained enough to support rigorous proofs.

The results are as follows. First, for any predictable memory policy, the LC-ERW admits an exact martingale normalisation. Secondly, if the learned memory coefficient converges fast enough to a diffusive value $a < 1/2$, then the normalised position $S_n/\sqrt{n}$ converges to a centred normal distribution with variance $(1-2a)^{-1}$. Thirdly, a corollary gives a direct stability condition for online-learning policies: any convergent policy satisfying a mild summability condition inherits the diffusive ERW limit theorem. The numerical section supplies reproducible R code that checks the predicted variance.

The manuscript is not a claim of peer-reviewed novelty. Rather, it states a precise candidate contribution, proves it under transparent assumptions and provides scripts that make the quantitative claims testable.

2 Related Literature

Schütz and Trimper (2004) introduced the ERW as a solvable non-Markovian model with long-range memory. Limit theorems and related asymptotic properties were developed by Coletti et al. (2017), while Bercu (2018) showed that martingale methods give a particularly effective route to strong laws, quadratic strong laws, laws of the iterated logarithm and non-Gaussian behaviour in the superdiffusive regime.

The connection with urn models was clarified by Baur and Bertoin (2016), and extensions have since become an active topic. Gut and Stadtmüller (2021) and Gut and Stadtmüller (2023) discuss variants with modified memory mechanisms. Dedecker et al. (2023) study random step sizes and rates of convergence, while Aguech (2024b) and Aguech (2024a) examine gradually increasing memory, random step sizes and delays. Recent work on trained ERW models studies how growing initial conditioning changes return behaviour (Dean, 2026). These papers motivate the present study but do not, to the best of our knowledge, formulate the memory parameter as the output of a general predictable learning rule.

3 The Learning-Controlled Model

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with filtration $(\mathcal{F}_n)_{n\geq 1}$. The position of the walker after n steps is

$$S_n = \sum_{j=1}^{n} X_j, \qquad X_j \in \{-1, 1\}.$$

The first increment satisfies $\mathbb{P}(X_1 = 1) = q$ and $\mathbb{P}(X_1 = -1) = 1-q$, where $q \in [0,1]$.

For $n \geq 1$, let K_n be conditionally uniform on $\{1, \dots, n\}$ given $\mathcal{F}_n$, and let p_n be an $\mathcal{F}_n$-measurable random variable taking values in $(0,1)$. Conditional on $\mathcal{F}_n$ and K_n, define

$$X_{n+1} = \begin{cases} X_{K_n}, & \text{with probability } p_n, \\ -X_{K_n}, & \text{with probability } 1-p_n. \end{cases}$$

The process p_n is the learning-controlled memory policy. In AI language, it may be written as

$$p_n = \ell(\theta_n),$$

where θ_n is the state of an online learning algorithm and $\ell : \mathbb{R}^d \to (0,1)$ is a bounded policy link, for example a logistic map. Set

$$a_n = 2p_n - 1.$$

The coefficient $a_n \in (-1, 1)$ is the learned memory coefficient. Positive values favour repetition of the recalled step, negative values favour reversal, and $a_n = 0$ corresponds to no directional memory in conditional mean.

Assumption 1 (Stable diffusive learning)**.** *There exist constants $a \in (-1, 1/2)$, $\varepsilon \in (0,1)$ and $\bar{a} < 1/2$ such that, almost surely,*

$$a_n \in [-1+\varepsilon,\, 1-\varepsilon] \quad \textit{and} \quad a_n \le \bar{a} \quad \textit{for all } n,$$

and

$$\sum_{n=1}^{\infty} \frac{|a_n - a|}{n} < \infty.$$

Remark 1. The summability condition is satisfied whenever $a_n \to a$ at any polynomial rate, for instance $|a_n - a| = O(n^{-\rho})$ with $\rho > 0$. It is deliberately stated pathwise, because the learning rule may be deterministic, data-driven or generated by a stochastic approximation algorithm.

4 Main Results

Theorem 1 (Exact martingale normalisation)**.** *Define $A_1 = 1$ and, for $n \ge 1$,*

$$A_{n+1} = \prod_{k=1}^{n} \left(1 + \frac{a_k}{k}\right).$$

Then

$$M_n = \frac{S_n}{A_n}, \qquad n \ge 1,$$

is a martingale with respect to $(\mathcal{F}_n)$.

Proof. Since K_n is conditionally uniform on $\{1, \ldots, n\}$,

$$\mathbb{E}[X_{K_n} \mid \mathcal{F}_n] = \frac{1}{n} \sum_{j=1}^{n} X_j = \frac{S_n}{n}.$$

Using the conditional repeat–reverse rule,

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = (p_n - (1-p_n)) \frac{S_n}{n} = a_n \frac{S_n}{n}.$$

Therefore

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + a_n \frac{S_n}{n} = \left(1 + \frac{a_n}{n}\right) S_n.$$

By definition, $A_{n+1} = (1 + a_n/n) A_n$. Hence

$$\mathbb{E}\left[\frac{S_{n+1}}{A_{n+1}} \middle| \mathcal{F}_n\right] = \frac{S_n}{A_n} = M_n.$$

The increments are bounded after normalisation on every finite horizon, so integrability is immediate. □

Lemma 1 (Second moment control)**.** *Under Assumption 1, there is a deterministic constant C_2 such that*

$$\sup_{n \ge 1} \frac{\mathbb{E}[S_n^2]}{n} \le C_2.$$

Proof. Let $V_n = \mathbb{E}[S_n^2]$. Since $X_{n+1}^2 = 1$,

$$\mathbb{E}[S_{n+1}^2 \mid \mathcal{F}_n] = S_n^2 + 2 S_n \mathbb{E}[X_{n+1} \mid \mathcal{F}_n] + 1 = \left(1 + \frac{2a_n}{n}\right) S_n^2 + 1.$$

Taking expectations and using Assumption 1 gives, for all $n \geq 1$,

$$V_{n+1} \leq \left(1 + \frac{2\bar{a}}{n}\right) V_n + 1.$$

A standard comparison with the deterministic recursion $u_{n+1} = (1+2\bar{a}/n)u_n+1$ yields $u_n = O(n)$ when $2\bar{a} < 1$. Therefore $V_n = O(n)$, as required. $\square$

Theorem 2 (Diffusive central limit theorem)**.** *Under Assumption 1,*

$$\frac{S_n}{\sqrt{n}} \xrightarrow{\mathcal{D}} \mathcal{N}\left(0, \frac{1}{1-2a}\right).$$

Proof. Write the martingale increments as

$$\Delta M_{k+1} = M_{k+1} - M_k = \frac{X_{k+1} - a_k S_k/k}{A_{k+1}}.$$

The predictable quadratic variation is

$$\langle M \rangle_n = \sum_{k=1}^{n-1} \frac{1 - a_k^2 S_k^2/k^2}{A_{k+1}^2}.$$

Assumption 1 implies, by logarithmic expansion,

$$A_n = L\, n^a(1 + o(1))$$

for an almost surely finite and strictly positive random variable L. Indeed,

$$\log A_n = \sum_{k=1}^{n-1} \log\left(1 + \frac{a_k}{k}\right) = a \log n + O(1) + o(1),$$

because $\sum_k |a_k - a|/k < \infty$ and $\sum_k k^{-2} < \infty$.

It follows that

$$\frac{A_n^2}{n} \sum_{k=1}^{n-1} \frac{1}{A_{k+1}^2} \longrightarrow \frac{1}{1-2a}.$$

The correction term satisfies

$$\mathbb{E}\left[\frac{A_n^2}{n} \sum_{k=1}^{n-1} \frac{a_k^2 S_k^2/k^2}{A_{k+1}^2}\right] \leq C \frac{A_n^2}{n} \sum_{k=1}^{n-1} \frac{1}{A_{k+1}^2} \frac{\mathbb{E}[S_k^2]}{k^2}.$$

By Lemma 1, $\mathbb{E}[S_k^2]/k^2 = O(1/k)$. A Toeplitz/Kronecker argument then gives convergence of this term to zero; hence

$$\frac{A_n^2}{n} \langle M \rangle_n \xrightarrow{\mathbb{P}} \frac{1}{1-2a}.$$

For the martingale Lindeberg condition, observe that

$$|\Delta M_{k+1}| \leq \frac{2}{A_{k+1}}.$$

Because $A_k \asymp k^a$ and $a < 1/2$, the largest martingale increment is negligible relative to $\langle M \rangle_n^{1/2}$:

$$\max_{k \leq n} \frac{|\Delta M_k|}{\langle M \rangle_n^{1/2}} \xrightarrow{\mathbb{P}} 0.$$

Thus the conditional Lindeberg condition holds. The martingale central limit theorem (see Hall and Heyde, 1980) gives

$$\frac{M_n}{\langle M \rangle_n^{1/2}} \xrightarrow{\mathcal{D}} \mathcal{N}(0,1).$$

Combining this with the convergence of $(A_n^2/n)\langle M \rangle_n$ and Slutsky's theorem yields

$$\frac{S_n}{\sqrt{n}} \xrightarrow{\mathcal{D}} \mathcal{N}\left(0, \frac{1}{1-2a}\right).$$

□

Corollary 1 (Stability of convergent learning policies)**.** *Let $\theta_n \in \mathbb{R}^d$ be an $\mathcal{F}_n$-predictable learning state and let $\ell : \mathbb{R}^d \to (\varepsilon, 1-\varepsilon)$ be Lipschitz. Put*

$$p_n = \ell(\theta_n), \qquad a_n = 2p_n - 1.$$

If $\theta_n \to \theta_\star$ almost surely and

$$\sum_{n=1}^{\infty} \frac{\|\theta_n - \theta_\star\|}{n} < \infty,$$

with $a = 2\ell(\theta_\star) - 1 < 1/2$, and if there exists $\bar{a} < 1/2$ such that $2\ell(\theta_n) - 1 \leq \bar{a}$ for all n, then the LC-ERW satisfies the central limit theorem in Theorem 2.

Proof. By Lipschitz continuity,

$$|a_n - a| \leq 2\ \mathrm{Lip}(\ell)\|\theta_n - \theta_\star\|.$$

The displayed summability condition implies Assumption 1, and Theorem 2 applies. □

5 Examples of Learning Rules

Deterministic annealing. Let $p_n = p_\star + c(n+1)^{-\rho}$, truncated to $(\varepsilon, 1-\varepsilon)$, where $p_\star < 3/4$ and $\rho > 0$. Then $a_n \to a = 2p_\star - 1 < 1/2$ and the summability condition holds.

Online logistic policy. Let $p_n = \ell(\theta_n)$, where $\ell(x) = \varepsilon + (1-2\varepsilon)/(1+\exp(-x))$, and suppose θ_n is an online estimator converging to $\theta_\star$ at a polynomial rate. The corollary gives the limiting law whenever $\ell(\theta_\star) < 3/4$.

Adaptive exploration controller. Let an AI controller observe S_n/n and update

$$\theta_{n+1} = \theta_n + \gamma_{n+1}\{b - \theta_n - \lambda S_n/n\},$$

with $\gamma_n \asymp n^{-\eta}$, $\eta \in (1/2, 1]$, and a stable target b. Under standard stochastic approximation stability conditions, the controller converges to b, so the LC-ERW has the diffusive Gaussian limit with $a = 2\ell(b) - 1$, provided $a < 1/2$. Classical sufficient conditions for such recursions are given by Kushner and Yin (2003).

6 Numerical Experiments

The accompanying R scripts simulate the LC-ERW for several policies:

1. a fixed-memory benchmark;
2. a deterministic annealing policy;

3. a logistic learning policy with polynomial convergence;
4. a simple adaptive controller using the empirical drift S_n/n.

For each policy, the scripts estimate the empirical variance of $S_n/\sqrt{n}$, compare it with $(1-2a)^{-1}$, and report skewness, excess kurtosis and Kolmogorov–Smirnov diagnostics against the predicted Gaussian law. These computations do not prove the theorem; they check whether finite sample behaviour is consistent with the asymptotic prediction.

7 Discussion

The LC-ERW shows that an AI-style adaptive memory policy can be incorporated into an ERW without destroying its martingale backbone. The main theorem says that, in the diffusive regime, stable learning is asymptotically equivalent to freezing the memory parameter at its learned limit. This is useful in two directions. For probability theory, it extends ERW analysis from fixed or externally prescribed parameters to predictable data-driven policies. For AI, it supplies a rare case in which an adaptive exploration mechanism with complete memory has a transparent asymptotic law.

Several limitations remain. The theorem does not cover critical learning limits $a = 1/2$, superdiffusive limits $a > 1/2$, multidimensional ERW, or learning rules whose parameter does not converge. These cases are likely to require different normalisations and may lead to non-Gaussian limits. Another open direction is statistical inference: estimating the learned memory trajectory p_n from observed LC-ERW paths.

8 Conclusion

This paper formulated a learning-controlled elephant random walk and proved that predictable adaptive memory policies admit an exact martingale normalisation. Under a stable diffusive learning condition, the process obeys a central limit theorem with variance determined by the learned limiting memory coefficient. The result gives a rigorous and reproducible starting point for connecting elephant random walks with AI-controlled stochastic exploration.

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